TEXORA · LECTURE SHEET

Colorimetry and Color Specification System

From reflectance curve to a single pass/fail number — CIE XYZ, CIELAB, colour-difference formulas, Kubelka–Munk and the specification systems buyers actually use.

WP-4103 · COLOR PHYSICS · TOPIC 2 · SPECIALIZATION — WET PROCESSING

1. Why colour must be specified numerically

Words such as “slightly redder” cannot run a global supply chain. **Colorimetry** is the science of measuring colour as numbers, so that a standard approved in one country can be reproduced, measured and accepted in another without shipping opinions back and forth. A complete instrumental specification needs three ingredients (from Topic 1): the sample’s **spectral reflectance** $R(\lambda)$, a **standard illuminant** SPD, and a **standard observer** (Topic 3). Every system in this sheet is built from those three.

2. CIE 1931 XYZ — tristimulus values

The CIE 1931 system summarises a colour stimulus by three **tristimulus values** X, Y and Z, obtained by multiplying, wavelength by wavelength, the reflectance $R(\lambda)$, the illuminant SPD $S(\lambda)$ and the observer’s colour-matching functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$, then summing over the visible spectrum:

$$X = k \Sigma S(\lambda)\, R(\lambda)\, \bar{x}(\lambda)\, \Delta\lambda \quad \text{(similarly Y with } \bar{y}\text{, Z with } \bar{z}\text{)}$$

Y is special: $\bar{y}(\lambda)$ was chosen to match the luminous efficiency function, so Y also carries the **luminance** (brightness) information. A spectrophotometer and its software perform this summation automatically; the user chooses illuminant and observer (in textiles, normally **D65 / 10°**).

Chromaticity coordinates

XYZ mixes lightness with chromatic quality. Dividing each value by the total separates pure chromaticity from luminance:

$$x = X / (X + Y + Z) \quad y = Y / (X + Y + Z) \quad z = Z / (X + Y + Z), \quad \text{with } x + y + z = 1$$

WORKED EXAMPLE 1 — CHROMATICITY FROM XYZ

A sample under D65 gives X = 25.5, Y = 40.0, Z = 18.2.
Sum = 25.5 + 40.0 + 18.2 = 83.7.
x = 25.5 / 83.7 = **0.305**; y = 40.0 / 83.7 = **0.478**; z = 18.2 / 83.7 = **0.217**. Check: 0.305 + 0.478 + 0.217 = 1.000 ✓. Since only two coordinates are independent, the colour's chromaticity is plotted as the point (x = 0.305, y = 0.478) on the chromaticity diagram; Y = 40.0 is reported alongside for lightness.

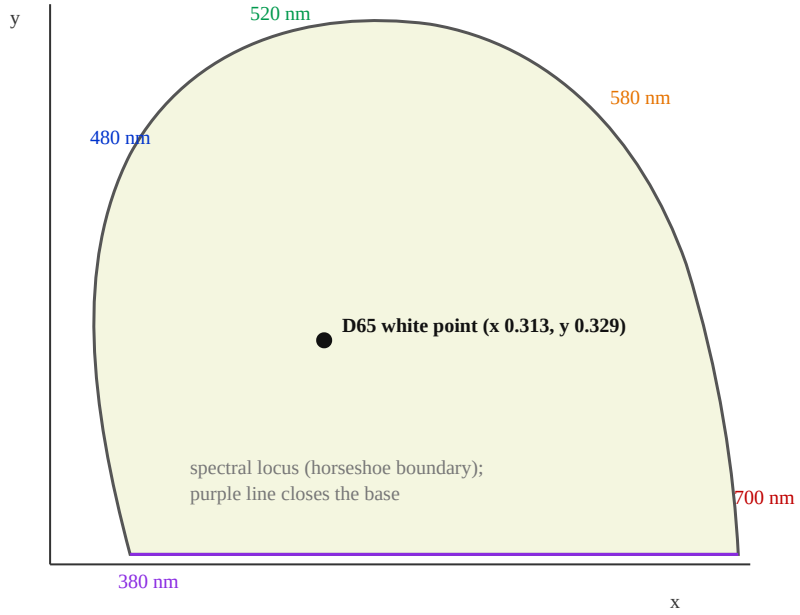


Figure 1 — CIE chromaticity diagram (sketch): the horseshoe spectral locus, the purple line, and the D65 white point.

On the **chromaticity diagram**, the horseshoe boundary is the *spectral locus* — the purest spectral colours — and all real colours fall inside it. The diagram is excellent for showing hue and for metamerism studies, but it is perceptually non-uniform (equal distances are not equal visual differences, notably in the greens) and carries no lightness. Those defects motivate CIELAB.

3. CIE 1976 L*a*b* and L*C*h

CIELAB transforms XYZ (normalised by the white point $X_nY_nZ_n$) through cube-root functions into a space designed to be approximately perceptually uniform, organised as opponent channels (Topic 3):

- **L*** — lightness, 0 (black) to 100 (white).
- **a*** — red–green axis: positive = redder, negative = greener.
- **b*** — yellow–blue axis: positive = yellower, negative = bluer.

The cylindrical form **L*C*h** re-expresses the same point in dyers' language: **C*** = $\sqrt{(a^{*2} + b^{*2})}$ is **chroma** (saturation/strength of colour) and **h** = $\tan^{-1}(b^*/a^*)$ is the **hue angle** (0° = red, 90° = yellow,

180° = green, 270° = blue). Production staff usually reason in L*C*h — “lighter, duller, same hue” — while instruments store L*a*b*.

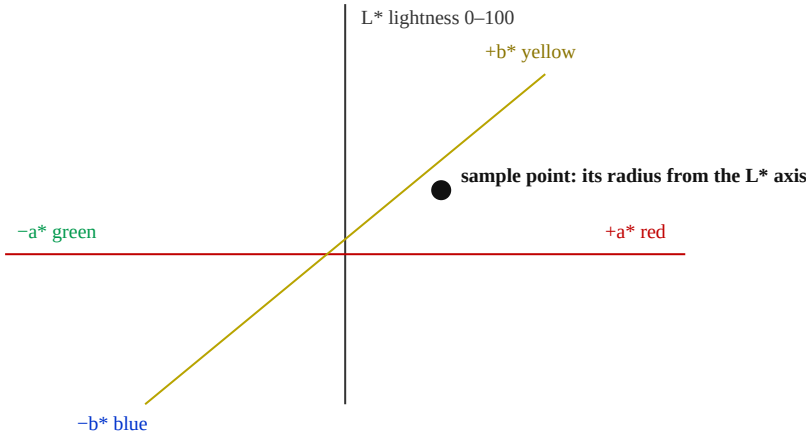


Figure 2 — CIELAB space (sketch): L* vertical axis with the a* and b* opponent axes; C* and h are the polar form of the a*–b* position.

4. Colour difference — ΔE*ab and its components

The colour difference between batch and standard in CIELAB is the straight-line (Euclidean) distance:

$$\Delta E^*_{ab} = \sqrt{(\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2})} = \sqrt{(\Delta L^{*2} + \Delta C^{*2} + \Delta H^{*2})}$$

The components diagnose the fault, which a single ΔE cannot: **ΔL*** lightness difference (batch lighter if positive), **ΔC*** chroma difference (duller if negative), and **ΔH*** hue difference, obtained from $\Delta H^* = \sqrt{(\Delta E^{*2} - \Delta L^{*2} - \Delta C^{*2})}$ with its sign assigned from the direction of hue rotation.

WORKED EXAMPLE 2 — FULL COLOUR-DIFFERENCE DIAGNOSIS (D65/10°)

Standard: L* 50.20, a* −15.30, b* 22.40. Batch: L* 51.05, a* −14.10, b* 20.85.

ΔL* = 51.05 − 50.20 = **+0.85**; Δa* = −14.10 − (−15.30) = **+1.20**; Δb* = 20.85 − 22.40 = **−1.55**.

$\Delta E^*_{ab} = \sqrt{(0.85^2 + 1.20^2 + 1.55^2)} = \sqrt{(0.7225 + 1.44 + 2.4025)} = \sqrt{4.565} =$ **2.14**.

$C^*_{std} = \sqrt{(15.30^2 + 22.40^2)} = \sqrt{735.85} =$ **27.13**; $C^*_{batch} = \sqrt{(14.10^2 + 20.85^2)} = \sqrt{633.53} =$ **25.17**;

ΔC* = 25.17 − 27.13 = **−1.96**.

$\Delta H^* = \sqrt{(2.14^2 - 0.85^2 - 1.96^2)} \approx$ **0.12** — negligible.

Diagnosis: the batch is lighter, less chromatic (duller) and almost identical in hue — mostly a chroma loss, i.e. a depth/strength problem, not a hue problem. Against a typical tolerance of ΔE ≤ 1.0 it **fails**; the correction is dye addition for depth, not a hue adjustment.

5. Advanced formulas — CMC and CIEDE2000

CIELAB is only approximately uniform: the eye tolerates larger differences in saturated colours and in lightness than ΔE^*_{ab} suggests. Two weighted systems correct this and dominate textile practice:

FORMULA	IDEA	ROLE IN TEXTILES
CMC(l:c)	Ellipsoidal tolerance around the standard; separate lightness (l) and chroma (c) weighting, commonly 2:1 for acceptability	Long-standing textile industry standard; a CMC(2:1) value of 1.0 is the classic pass/fail boundary in many buyer specifications
CIEDE2000 (ΔE_{00})	Further weighting functions for lightness, chroma and hue plus a rotation term for the blue region	Current CIE recommendation; best agreement with visual assessments; increasingly specified by retailers. Its full computation is lengthy and is always performed by software

The working principle across all three formulas is identical: specify the formula, the illuminant/observer and the tolerance together — a "ΔE of 0.9" is meaningless until all three are stated.

6. Kubelka–Munk — linking colour to dye concentration

For opaque, light-scattering materials such as textiles, the **Kubelka–Munk** relation converts reflectance at a wavelength into a quantity proportional to colorant concentration:

$$K/S = (1 - R)^2 / 2R$$

where K is absorption and S scattering. For a single dye on a given substrate, K/S at the absorption peak rises linearly with dye concentration — the basis of **computer colour matching**: the K/S curves of the component dyes add in a mixture, so software can predict the recipe that reproduces a target reflectance curve.

WORKED EXAMPLE 3 — KUBELKA–MUNK VALUES, INVERSE AND A MIXTURE

R = 0.20 → K/S = (0.80)² / 0.40 = 0.64 / 0.40 = **1.60**.

R = 0.50 → K/S = (0.50)² / 1.00 = **0.25**. R = 0.08 → K/S = (0.92)² / 0.16 = 0.8464 / 0.16 = **5.29**.

Note how steeply K/S climbs as reflectance falls — deep shades are K/S-sensitive.

Inverse: K/S = 1.60 returns R = **0.20** (the relation is single-valued for opaque samples).

Mixture additivity at one wavelength: 0.6 unit of dye A (K/S 2.4) + 0.4 unit of dye B (K/S 1.1) → K/S = 0.6 × 2.4 + 0.4 × 1.1 = 1.44 + 0.44 = **1.88**.

Limitations: Kubelka–Munk assumes an opaque, homogeneous, diffusely scattering layer. Fluorescent samples, very pale shades (substrate dominates), lustrous surfaces and mixtures on a different substrate

all degrade its accuracy, which is why match-prediction software is calibrated with dyeings on the actual production substrate.

7. Specification systems — Munsell and Pantone

Not all specification is instrumental XYZ arithmetic. Two physical/visual systems remain in wide commercial use:

SYSTEM	ORGANISATION	NOTATION / FORM	USE
Munsell	Colour ordered by perceived hue, value (lightness) and chroma, as painted chips in an atlas	e.g. 5R 4/6 = hue 5R, value 4, chroma 6	Perceptually even reference; education, visual specification, interpolation between chips
Pantone (Fashion, Home + Interiors / TCX)	Proprietary swatch library of dyed cotton standards with codes	e.g. Pantone 19-4052 TCX “Classic Blue”	The fashion industry’s common language: buyers nominate a TCX code as the shade standard; the laboratory then matches it instrumentally

Physical standards fade, soil and vary between editions, so professional practice treats the swatch or code as the *reference* and the measured L*a*b* values (D65/10°) as the *working specification* that travels between mills.

8. Tolerances and pass/fail

A **tolerance** is the largest permitted difference from the standard, set in a stated formula: for example “CMC(2:1) ≤ 1.0 under D65/10°, plus ΔL* within ±0.5”. One-number ΔE tolerances are easy to abuse because ΔE hides direction: two batches can share ΔE = 0.9, one too red and one too dull, and only component limits (ΔL*, ΔC*, ΔH*) catch commercial drift. Sound specifications therefore combine: (1) a ΔE formula and limit; (2) component limits for production control; (3) a metamerism index or second-illuminant check (A or TL84); and (4) a visual assessment in a light cabinet as the final arbiter for borderline cases. Worked Example 2 shows the routine: ΔE 2.14 against a 1.0 tolerance = fail, and the components tell the dyer exactly what to correct.

9. Glossary — key terms

- **Tristimulus values (XYZ):** three numbers summarising a colour stimulus for a stated illuminant and observer.

- **Chromaticity coordinates (x, y):** normalised XYZ locating chromaticity independent of luminance.
- **CIELAB (L*a*b*):** approximately uniform opponent colour space.
- **Chroma (C*) / hue angle (h):** polar form of a*, b* — saturation and hue.
- **ΔE:** total colour difference in a stated formula; ΔL*, ΔC*, ΔH* are its diagnostic components.
- **K/S:** Kubelka–Munk absorption/scattering ratio, proportional to colorant concentration.
- **Tolerance:** the contractual maximum permitted difference from the standard.

10. Key points

- Every colour number requires illuminant + observer + formula to mean anything; textiles default to D65/10°.
- Diagnose with components, decide with ΔE: ΔL*, ΔC*, ΔH* show lightness, chroma and hue faults separately.
- CMC(2:1) and CIEDE2000 correct CIELAB non-uniformity and are the buyer-facing formulas.
- K/S additivity is the engine of computer colour matching; Munsell and Pantone TCX remain the visual specification languages.

11. Summary

Colorimetry converts a reflectance curve, under a standard illuminant and observer, into XYZ; chromaticity separates colour quality from luminance, and CIELAB reorganises the result into the lightness–chroma–hue terms in which dyers think and buyers set tolerances. Weighted formulas (CMC, CIEDE2000) align the numbers with the eye, Kubelka–Munk ties them back to dye on fibre, and specification systems such as Pantone TCX give the whole chain a shared standard to aim at.

12. Practice

1. X = 30.0, Y = 25.0, Z = 12.0. Calculate x, y and z and verify they sum to 1.
2. Standard L* 60.0, a* 20.0, b* 10.0; batch L* 59.2, a* 21.1, b* 9.4. Calculate ΔL*, Δa*, Δb* and ΔE^*_{ab}, and state the batch's faults in words.
3. Calculate K/S for R = 0.35, and the K/S of a mixture of 0.7 unit of dye P (K/S 1.8) with 0.3 unit of dye Q (K/S 0.9) at the same wavelength.
4. Why must a buyer's tolerance state the formula (e.g. CMC(2:1)) and not merely "ΔE ≤ 1.0"?
5. A batch passes ΔE but is consistently too dull across repeats. Which component limit should the specification have included?

Answer key

1. Sum = 67.0. x = 30.0/67.0 = 0.448; y = 25.0/67.0 = 0.373; z = 12.0/67.0 = 0.179. Total = 1.000 ✓.
2. ΔL* = −0.8, Δa* = +1.1, Δb* = −0.6. ΔE^*_{ab} = √(0.64 + 1.21 + 0.36) = √2.21 = 1.49. The batch is darker, redder and slightly bluer (less yellow) than standard.
3. K/S = $(0.65)^2$ / 0.70 = 0.4225 / 0.70 = 0.60. Mixture: 0.7 × 1.8 + 0.3 × 0.9 = 1.26 + 0.27 = 1.53.
4. Because the formulas weight lightness, chroma and hue differently, the same pair of samples gives different ΔE values in ΔE^*_{ab}, CMC and CIEDE2000; without the named formula (and illuminant/observer) the limit is unenforceable.
5. A ΔC* limit (minimum permitted chroma difference band, e.g. ΔC* within −0.5), so dullness is caught even when total ΔE stays inside tolerance.

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